

Synthesis of Paracetamol through Free-Radical Reactions: An Educational Approach in Organic Chemistry

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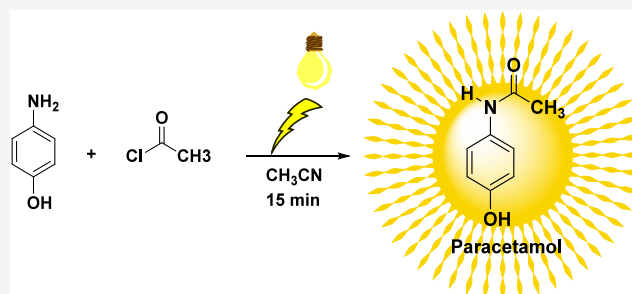
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ABSTRACT: Acquiring the skills to improve synthetic techniques used in research and industry is one of the most critical goals for chemists in training. Achieving this requires not only a solid foundation in chemistry but also creativity and a deep understanding of the physicochemical processes involved in experimental work. This experimental activity is designed to guide students in developing a novel synthetic technique for producing paracetamol, which is a widely used pharmaceutical compound. Students are encouraged to first rationalize the key steps in the currently used procedures, identifying potential limitations and areas for improvement. This laboratory synthesis is specifically intended for second-semester organic chemistry students. Through bibliographic research and group brainstorming, they then proposed a new reaction pathway based on free-radical chemistry. Using their acquired knowledge and the transition wavelengths of the compounds involved, students design an appropriate experimental setup based on irradiation of electromagnetic waves in the UV region. This innovative approach employs commercial reagents, such as 4-aminophenol and acetyl chloride, and achieves a reaction time of just 15 min with yields exceeding 70%, outperforming other synthetic methods reported in the literature. The purity and chemical structure of the synthesized paracetamol were confirmed through ^1H nuclear magnetic resonance (NMR) spectroscopy, mass spectrometry (DART method), and melting point analysis.

KEYWORDS: Paracetamol, Free radicals, Pharmaceutical Engineering Degree, Organic Chemistry Laboratory



INTRODUCTION

The adaptation of new synthetic techniques for producing widely used chemical compounds, especially those with applications in human health, has profound implications for both economic growth and societal well-being. As future chemists, bachelor's students must develop the skills necessary to propose innovative pathways with potential industrial relevance. This requires fostering initiative, creativity, and a solid foundation of fundamental chemical knowledge to design plausible strategies that can then be validated through laboratory experimentation.

In modern organic chemistry, various synthetic techniques are actively explored with the aim of revitalizing traditional methods and promoting more sustainable and didactic practices. Among these, multicomponent reaction (MCR) strategies stand out for their efficiency and versatility in obtaining high value-added products, as has been implemented in laboratory experiments for undergraduate students.^{1–5} Likewise, ultrasound-assisted methods have become effective tools for accelerating reactions and enabling milder reaction conditions, which have been successfully incorporated into organic chemistry laboratory practices for second-semester organic chemistry students.^{6–8} These methodologies comple-

ment and enrich the teaching of organic synthesis, aligning with the development of skills in experimental design and critical thinking. In this context, in a recent study, we successfully synthesized moclobemide, an important anxiolytic, within just 5 min, using a UV lamp through a free-radical mechanism, thus demonstrating the efficiency and potential of these new methodologies in drug synthesis.⁹

In this context, this experimental activity was designed with a dual purpose: on one hand, to update and optimize the synthetic pathway for producing one of the most widely used analgesics, paracetamol (acetaminophen, Figure 1), through free-radical generation induced by UV irradiation; and on the other hand, to strengthen students' understanding of reaction mechanisms and radical processes in a practical and engaging setting. Paracetamol **1** was first synthesized by the American chemist Harmon Northrop Morse in 1878; however, its initial

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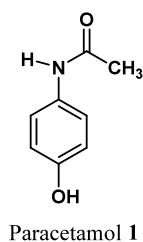
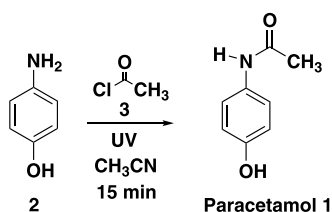


Figure 1. Chemical structure of paracetamol (acetaminophen).

production was not intended for medicinal purposes but rather was part of basic research aimed at producing phenolic derivatives. Morse obtained it from *p*-aminophenylacetic acid. Although this early synthesis did not have an immediate pharmaceutical application, it laid the groundwork for the later development of modern paracetamol production.^{9,10} It was not until the 1940s that paracetamol began to be widely used as an analgesic and antipyretic, following confirmation of its safety and effectiveness for relieving pain and fever. Today, it is effective for treating various types of mild to moderate pain such as headaches, muscle aches, back pain, menstrual cramps, and toothaches as well as for reducing fever associated with colds and flu.^{11–13} However, it should be used responsibly and according to recommended dosages, as an overdose can cause severe liver damage.¹⁴

The proposed strategy, outlined in Scheme 1, is both straightforward and efficient. It employs commercially available

Scheme 1. Proposed Synthesis of Paracetamol



reagents 4-aminophenol 2 and acetyl chloride 3 under standard reaction conditions alongside a UV lamp to initiate the formation of highly reactive radical intermediates. These intermediates rapidly convert into paracetamol, offering a synthetic route that is more selective and efficient than other methods.^{15–17} Paracetamol is obtained in just 15 min with a

yield of 70%, while the remainder consists of unreacted 4-aminophenol 2, which can be recovered and reused. This renewed procedure significantly reduces reaction times, eliminates the need for sophisticated equipment and minimizes the environmental impact of the paracetamol production chain.

Moreover, it serves as an accessible and high-value educational experiment, integrating principles of sustainability and mechanistic understanding, while fostering students' skills in experimental design and critical thinking.

Comparison between Mechanistic Pathways

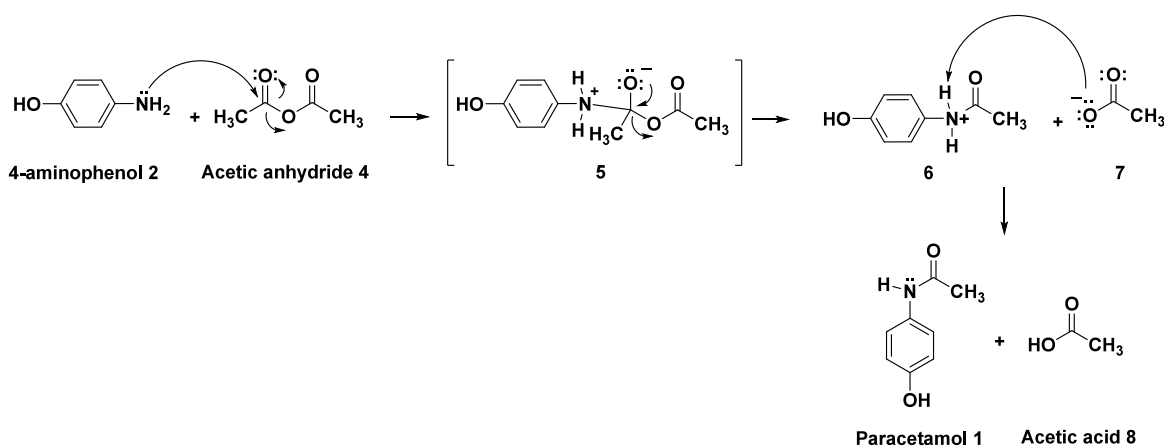
To enhance the flux of discussion, in this section, we present the basics of the two mechanistic pathways presented to students for the synthesis of acetaminophen, starting from *p*-aminophenol.

The “standard” procedure (Scheme 2), the synthesis of paracetamol proceeds through a straightforward acetylation reaction, initiated by the nucleophilic attack of the amino group of 4-aminophenol 2 on acetic anhydride 4. In this step, the lone pair on the nitrogen atom attacks the carbonyl carbon of the anhydride, leading to the formation of amide intermediate 5. Subsequently, the lone pair on the oxygen atom within the intermediate facilitates an intramolecular rearrangement, which generates a pair of ionic species: a protonated amide ion (6) and an acetate ion (7). The negatively charged oxygen of the acetate ion then acts as a base, abstracting a proton from the protonated amide ion. This final proton transfer step yields the target compound paracetamol 1, along with acetic acid 8 as a byproduct. It is important to highlight that, in most syntheses, following this procedure, acetic acid is not recovered and is often discarded into the environment, contributing to pollution.

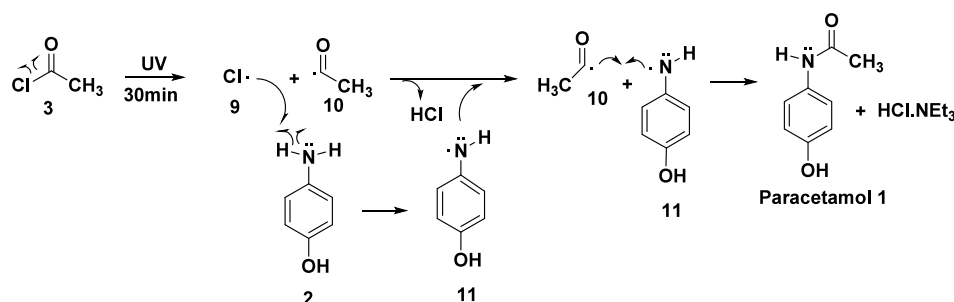
The authors tentatively suggest the mechanism depicted in Scheme 3 for the synthetic procedure developed in this work, which can be summarized as follows: acetyl chloride 3 undergoes homolytic cleavage, generating two radical species, 9 and 10. Radical 9 subtracts a hydrogen atom from 4-aminophenol 2, forming radical 11, releasing HCl, which is subsequently neutralized by NEt₃. Following this, radical 10 forms a covalent bond with radical 11, ultimately yielding paracetamol 1.

To elucidate the reaction mechanism of acetaminophen synthesis mediated by free radicals, it is essential to consider key aspects of bond chemistry, particularly the behavior of acyl–Cl bonds in compounds such as acetyl chloride.

Scheme 2. Standard Mechanistic Pathway for the Synthesis of Paracetamol



Scheme 3. Reaction Mechanism of Paracetamol via Free Radicals



According to Arunan, the bond dissociation energy (BDE) for the acyl–Cl bond in acetyl chloride is 83 kcal/mol, while Ewing indicates that the dissociation of the C–Cl bond primarily occurs when irradiated at 193 nm.^{18,19} This relatively low BDE suggests that cleavage of the acyl–Cl bond can readily occur under specific conditions, making it more prone to homolytic fission compared to other bonds of similar nature. This ease of dissociation favors the generation of free radicals, which play a crucial role in initiating the reaction pathway, leading to the formation of acetaminophen. Based on this understanding, students are encouraged to intentionally modify the currently used synthetic procedure, exploring the new approach proposed, (Scheme 2) toward the new approach presented in Scheme 3.

Pedagogical Objectives

In this experimental activity, students explore how modifying reaction conditions can significantly optimize a synthetic procedure. To achieve this, they critically analyze and compare two different mechanistic pathways for synthesizing paracetamol: the traditional addition–elimination mechanism and an alternative free radical-mediated approach. This practice not only reinforces concepts related to standard analytical techniques but also has, as a fundamental pedagogical feature, the development of students' understanding of the use of free-radical mechanisms in organic synthesis. The following pedagogical objectives have been established for this experiment: Deepen the understanding of mechanistic pathways, focusing on addition–elimination processes and their significance in organic synthesis.

- Enhance knowledge of free-radical chemistry, particularly the reactivity of aniline and acyl chloride groups in radical-driven reactions.
- Introduce a simple and practical experimental procedure for synthesizing paracetamol, a widely used pharmaceutical compound.
- Identify the significance of photochemical activation as a tool for initiating free-radical reactions.
- Critically analyze and compare two distinct mechanistic pathways for synthesizing paracetamol: the addition–elimination mechanism and a free radical-mediated approach.
- Reinforce the use of standard analytical techniques, such as melting point determination, mass spectra and NMR spectroscopy, for characterizing synthetic products.
- Analyze and rationalize previous synthetic methods to identify possible areas for improvement, focusing on enhancing efficiency, reaction conditions and environment impact.

- Promote collaborative work and critical discussion, encouraging students to interpret data and propose improvements to the experimental design.
- Understand the relationship between the wavelength of electromagnetic radiation and the energy required for the dissociation of the C–Cl bond.

The pedagogical strategy employed in this work integrated both spectroscopic and classical characterization approaches, providing students with a robust framework for chemical analysis. The use of a 4-aminophenol reference spectrum offered a solid foundation for the development of skills in compound identification and interpretation of advanced spectroscopic data. In parallel, the students confirmed that the melting point of the synthesized paracetamol aligned with the values reported in the literature, a result that—although often underemphasized in academic curricula—proved to be of significant relevance. This characterization not only validated the purity of the compound but also reinforced the students' understanding of fundamental chemical principles and highlighted the importance of empirical validation in experimental practice. Beyond technical verification, this exercise underscored the transferability of knowledge: since the activity was grounded on a synthetic strategy previously developed by our group, it provided an excellent opportunity for students to appreciate how a novel yet well-established methodology can be adapted to address chemically meaningful problems with direct implications for the pharmaceutical and chemical industries. In doing so, the activity strengthened the students' ability to bridge theoretical concepts with practical applications, fostering a critical competency in modern chemical education.

Students were assigned to conduct a thorough literature search to determine which type of solvent is most suitable for such reactions. This exercise not only allowed them to develop critical research skills but also gave them the opportunity to apply theoretical knowledge in a practical context (Table 1).

Table 1. Selected Solvents in Order of Increasing Polarity

solvent	dielectric constant
hexanes	1.89
methylene chloride	8.93
acetone	20.7
ethanol	24.2
methanol	32.7
dimethyl sulfoxide (DMSO)	46.7

Learning Outcomes

- Fundamental principles of free radicals: Understand the basic concepts of free-radical chemistry, including their formation, stability, and reactivity, with a focus on their role in organic synthesis and in the functional groups involved in this proposal.
- Practical experience in the organic laboratory: Gain hands-on skills in conducting organic reactions, handling standard laboratory equipment, irradiating the reaction media with an UV-lamp and following safety protocols in a controlled environment.
- Synthesis of paracetamol: Perform the step-by-step synthesis of paracetamol, a widely used pharmaceutical compound, using both traditional and innovative methods.
- Mechanism of paracetamol synthesis via free radicals: Explore the detailed reaction mechanism involving free radicals, including the homolytic cleavage of acyl-Cl bonds and the formation of intermediates.
- Proper solvent mixtures: Under polarity arguments, select the most suitable binary solvent mixture for the purification of the paracetamol to be obtained. The considered solvents are listed in Table 1.
- Reaction progression by TLC: Use thin-layer chromatography (TLC) to monitor the progress of the reaction, identify intermediates, and confirm the completion of the synthesis. Students will also identify the adequate mobile phase for proper purification of paracetamol.
- Results analysis: Interpret experimental data to evaluate the success of the synthesis, including yield, purity, and efficiency of the reaction.
- Identification of paracetamol by NMR ^1H : Analyze and interpret ^1H NMR spectra that will be provided by the instructor, to confirm the chemical structure and purity of the synthesized paracetamol.
- Analysis and identification of paracetamol by mass spectrometry (DART): Use direct analysis in real-time (DART) mass spectrometry to characterize the molecular weight and structure of paracetamol.

Discussion and Activities Prior Synthesis of Paracetamol via Free Radicals

To help the students understand free-radical reactions, a theoretical class was offered prior to the synthesis. In this session, the three fundamental stages of a free-radical reaction were covered: initiation, propagation, and termination. Additionally, possible secondary products that could be produced during the reaction were discussed alongside the necessary precautions when working with free radicals, especially regarding the use of UV lamps.

Additionally, the difference between polar processes and free radical reactions was emphasized, the latter being characterized by homolytic cleavages under UV light, which demonstrates that the mechanism is different from that of a Lewis acid–base reaction. We explained to students that initiating these reactions relies on the absorption of energy from irradiated electromagnetic waves by the reactants. This energy absorption promotes the homolytic cleavage of chemical bonds, generating reactive intermediates such as free radicals.^{20–22} Importantly, each compound has specific wavelengths at which it optimally absorbs energy, dictated by its electronic structure and functional groups. Consequently, each reaction may have an optimal wavelength that maximizes the efficiency and

selectivity. Determining this wavelength is a critical step that must be established through careful analysis and experimentation prior to the conduct of the reaction.

The discussion focused on the key differences between the addition–elimination mechanism and the free radical-driven approach. In the addition–elimination method, the amino group of 4-aminophenol reacts selectively with acetyl chloride, ensuring a straightforward pathway to the desired product. In contrast, the free radical-driven mechanism can generate multiple intermediates, potentially reducing selectivity and leading to byproducts; nonetheless, students are advised that the new radical-based procedure does not provide undesired byproducts. Additionally, the reaction conditions differ significantly: the free-radical approach relies on energy absorption from electromagnetic waves (e.g., UV light), offering a controlled and energy-efficient process, while the addition–elimination method requires heating the reaction mixture, which involves more experimental precautions and longer reaction times and often results in lower yields. These distinctions underscore the advantages of the free-radical-driven method in terms of efficiency and experimental simplicity, despite its challenges with selectivity. It is important to note that students exclusively carried out the free-radical-based synthesis, whereas the addition–elimination mechanism was introduced solely for comparative and theoretical purposes.

As a part of the laboratory practices (a 3 h session), the students conducted experiments to determine the best conditions to achieve optimal performance in a chemical reaction. Variables, such as temperature and reaction time, were evaluated in an attempt to identify the conditions that would promote the best efficiency in the process, (Table 2). It

Table 2. Reaction Yields as a Function of Time for the Addition–Elimination (1–3) and free Radical-Driven (4–5) Mechanistic Pathways

reaction	temperature (°C)	time (min)	solvent	yield	$h\nu$
1	25	30	CH_3CN	30%	×
2	30	60	CH_3CN	50%	×
3	Reflux/82	30	CH_3CN	degradation	×
4	25	15	CH_3CN	70%	✓
5	25	30	CH_3CN	70%	✓

is worth emphasizing that the reactions presented in Table 2 were previously conducted by the instructors and used as illustrative examples to support and enhance student discussion. Students noted that these variations did not translate into a substantial improvement of the reaction efficiency. The obtained results, together with the previous discussion and revised bibliography, led the students to recognize that a different experimental protocol based on electromagnetic irradiation in the UV region has the potential to provide marked enhancement in the reaction yield. This exercise not only allowed the students to manipulate different experimental variables to improve reactions efficiency but also encouraged them to reflect on the importance of optimizing experimental conditions in scientific research and chemistry education.

Preparing the Experimental Setting

Using the gathered information, the group conducted a brainstorming session to develop a plausible proposal for

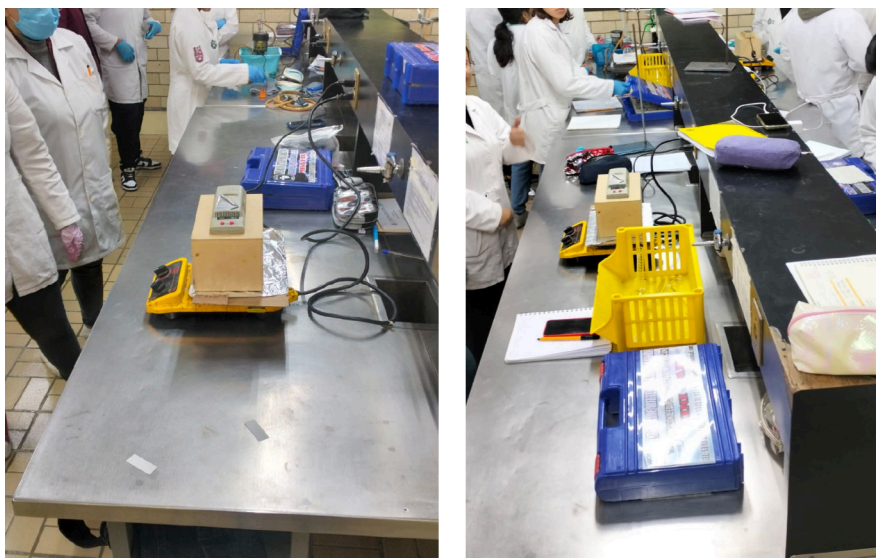


Figure 2. Experimental setup, including the UV irradiation device.

optimizing the synthetic procedure of paracetamol. They identified free radical generation via electromagnetic irradiation as a promising starting point given its potential to enhance efficiency and simplify experimental conditions. To maintain experimental simplicity and minimize energy waste, they decided to conduct the reaction at room temperature. Two critical aspects were then addressed: solvent selection and irradiation wavelength.

For the solvent, students were tasked with investigating the most suitable medium for free radical generation, which is a key step in photochemical reactions. After reviewing several scientific articles, they identified acetonitrile (MeCN) as an optimal choice due to its ability to dissolve a wide range of organic compounds, its stability under UV irradiation, and its favorable properties for molecular dissociation, which promote free radical formation.^{23–26}

Regarding the irradiation wavelength, students analyzed the UV absorption profiles of *p*-aminophenol and acetyl chloride. *p*-aminophenol exhibits strong absorption at 200–250 nm ($\pi \rightarrow \pi^*$ transitions) and weak absorption at 250–350 nm ($n \rightarrow \pi^*$ transitions), while acetyl chloride shows strong absorption at 200–220 nm ($\pi \rightarrow \pi^*$ transitions) and weak absorption at 280–300 nm ($n \rightarrow \pi^*$ transitions). It is pertinent to mention that we did not deepen the nature of these transitions as only the favorable absorption regions of the involved compounds are of interest for this experimental activity. Although the overlapping strong absorption region for both compounds lies between 200–220 nm, the instructor explained that 250 nm is a practical choice for UV lamps commonly used in teaching laboratories. This wavelength not only aligns with the strong absorption of *p*-aminophenol but also allows for the weak transitions of acetyl chloride, facilitating the homolytic cleavage of the C–Cl bond and free radical generation. Students were intrigued by the potential of 220 nm irradiation, as it could enhance free-radical formation for both reactants, but they ultimately agreed that 250 nm was a suitable and accessible option for the experiment in a teaching laboratory.

Experimental Procedure in the Teaching Laboratory

The experimental activity was carried out with a total of 60 students, divided into two groups of 30. Each group was organized into small teams to perform the synthesis of

paracetamol by using UV irradiation. The procedure began with the addition of the reagents (4-aminophenol and acetyl chloride) into a glass crystallizing dish measuring 6.5 cm $\times$ 2 cm, which was placed on a magnetic stirrer. A wooden box was then used to hold the UV lamp directly above the reaction mixture, allowing for controlled irradiation for 15 min (Figure 2). After the reaction was completed, students proceeded to purify the solid product and characterize it by measuring its melting point. They were also provided with NMR spectra (^1H and ^{13}C) and mass spectrometry data (DART). Each team prepared a technical report in which they justified the proposed reaction mechanism, interpreted the spectroscopic data, and discussed the efficiency of the methodology compared to traditional synthetic routes.

The students were instructed to monitor the progress of the reaction using thin-layer chromatography (TLC) in order to determine whether the reaction had been completed or whether any starting material remained. Each team was provided with a chromatographic plate for analysis. The mobile phase used was a 3:1 mixture of hexane and ethyl acetate, which yielded an $R_f = 0.28$ for paracetamol and a $R_f = 0.46$ for 4-aminophenol. The students carried out and analyzed the corresponding TLC plate, which is shown in Figure 3.

Purification. With the results obtained using thin-layer chromatography (TLC) in hand, the students purified the paracetamol using a mobile phase constituted by hexane and ethyl acetate in a 3:1 ratio (Figure 4).

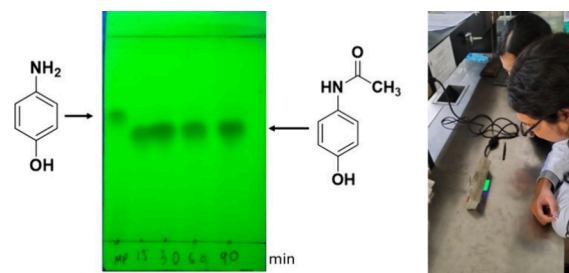


Figure 3. TLC chromatographic plate with varying times.



Figure 4. Chromatographic column used for the purification of paracetamol 1.

At the conclusion of the experimental session, the instructor provided students with a set of NMR spectra including only one spectrum corresponding to the synthesized compound. This spectrum was presented without prior elucidation, encouraging students to independently analyze and interpret the data. This approach aims to deepen students' understanding of spectroscopic characterization and develop their skills in identifying and confirming chemical structures using NMR techniques. It is pertinent to mention that all of the NMR measurements were exclusively carried out by the instructors.

Experimental Report Guidelines

At the end of the lab session, students are required to prepare and submit a comprehensive report. The report should include, but is not limited to, the following sections:

- (1) **Proposed Reaction Mechanism:** Provide a detailed step-by-step mechanism for the reaction performed during the lab session, including all intermediates and analyzing possible transition states.
- (2) **Comparison of Reaction Mechanisms:** Analyze and contrast the addition–elimination mechanism reported in the literature with the free-radical-driven mechanism employed in the experiment. Highlight the key differences in reactivity, selectivity, and efficiency.
- (3) **NMR Spectrum Analysis:** Perform a partial or complete elucidation of the ^1H NMR spectra provided at the end of the session (available in the [Supporting Information](#)). Assign peaks to specific protons and carbons in the product structure and discuss any notable features or anomalies.
- (4) **Mass Spectrometry Analysis:** Interpret the mass spectrometry data (provided at the end of the session and available in the [Supporting Information](#)). Identify the molecular ion peak, fragmentation patterns, and any evidence of byproducts or impurities.
- (5) **Melting Point Comparison:** Compare the experimentally determined melting point of the synthesized product with literature values. Discuss any discrepancies and their potential causes, such as impurities or experimental error. In this educational context, melting point determination serves not only as a means of assessing

sample purity but also as a complementary technique to molecular spectroscopy. Its inclusion in the experiment reinforces the concept of substance purity and provides students with a tangible, accessible method for characterizing chemical compounds.

- (6) **Byproduct Formation:** Analyze the possible byproducts that could form during the reaction. Emphasize how the chosen synthetic method minimizes byproduct formation and contributes to the overall efficiency, convenience, and environmental sustainability of the process.

Students are expected to submit this report one week after completing the experimental work, during the next lab session. The report should demonstrate critical thinking, clear reasoning, and a thorough understanding of the experimental and analytical techniques used.

EXPERIMENTAL DETAILS

Experimental Setup

All reagents were purchased from Sigma–Aldrich and used as received. Flash column chromatography was performed using Aldrich silica gel 230–400 mesh; 25 g of silica was used. ^1H NMR data for all synthesized compounds were recorded at room temperature using Bruker BioSpin GmbH (500 MHz, Switzerland) spectrometers. Our NMR experiments were carried out using deuterated chloroform (CDCl_3) as the solvent. Chemical shifts (δ) are reported in parts per million (ppm). Multiplicities were recorded as singlet (s), double–double (dd), and wide signal (bs). Coupling constants (J) are given in units of Hz. Melting points were determined using a Fisher Johns melting point apparatus (uncorrected). High-resolution mass spectra were recorded on a JEOL AccuTOF JMS-T100LC mass spectrometer with TOF mass analyzer. The lamp used was the UVGL-25 model, known as the Mineralight Lamp, which offers multiband functionality with wavelengths of 254/366 nm. This lamp operates at 115 V, 60 Hz, and 0.6 A.

Synthesis of Paracetamol 1

In a Petri dish measuring 6 cm in diameter and 3 cm in depth, equipped with a magnetic stir bar, 4-aminophenol (1 g, 1 equiv), acetyl chloride (0.72 mL, 1 equiv), and triethylamine (1.53 mL, 1.2 equiv) were added, along with 2 mL of acetonitrile. The Petri dish was placed on a stirring plate and covered with a wooden box designed to hold a UV lamp that emits light at 254 nm. The mixture was slowly stirred for 15 min under UV irradiation (see [Figures 5](#) and [6](#)). After this time, all volatile components were removed under a vacuum. The resulting product was purified by flash column



Figure 5. Paracetamol reaction.



Figure 6. Reaction with UV lamp.

chromatography using a 3:1 mixture of hexane and ethyl acetate (AcOEt) as the eluent.

Characterization of Paracetamol

Students elucidated the ^1H NMR spectrum obtained from Paracetamol. Yield using UV lamp: 0.95 g, (70%); solid white; mp = 170 °C (literature. m.p.: 170 °C); Rf = 0.28 on silica gel (hexane: ethyl acetate 3:1); ^1H NMR (500 MHz, CDCl_3): δ 2.18 (s, 3H), 3.66 (sb, -OH), 6.81 (d, 2H, J = 9.0), 7.05 (sb, -NH), 7.35 (d, 2H, J = 9.0). HRMS (DART) $^+$: calcd for $\text{C}_8\text{H}_{10}\text{NO}_2$ [$M + \text{H}$] 152.0711, found.152.0715.

Characterization of Paracetamol by Melting Point

The characterization was also carried out through melting point analysis, which was crucial to confirm the identity and purity of the formed product. Students were explained that if the product were contaminated, the melting point could deviate from the value reported in the literature, indicating the presence of impurities (Figure 7).



Figure 7. Students measuring the melting point of paracetamol.

Hazards

The use of ultraviolet (UV) lamps in the laboratory is common in various experimental techniques such as the initiation of free-radical reactions. Below are some key precautions students should take when working with UV lamps: Eye protection: UV light is harmful to the eyes. Direct exposure can cause retinal damage and conjunctivitis. Skin protection: Direct exposure of the skin to UV rays can cause burns or long-term damage. 4-Aminophenol should be handled with care; always use gloves, safety glasses, and a laboratory coat. It is important to avoid direct contact with the skin and eyes and, in case of exposure,

wash with plenty of water. It should be handled in well-ventilated areas, preferably under a fume hood, and stored in a cool, dry place with proper labeling. Acetyl chloride must be handled with caution, strictly within a fume hood, with contact with skin and eyes. In case of exposure, the affected area should be washed immediately with plenty of water. Store it in a cool, dry place, properly labeled, and away from heat sources. Acetonitrile should be handled with care, avoiding inhalation of vapor and contact with the skin or eyes. Work in well-ventilated areas, preferably under a fume hood. Store it in a cool, dry place away from heat sources or incompatible materials, and label it properly.

Pedagogical Evaluation

The assessment of student learning focused on the development of key competencies: chemical reasoning, experimental design, data analysis, and collaborative work. Student performance was evaluated using rubrics applied to their laboratory work, the quality of their experimental design, their analysis of the results, and the final report. In addition, a brief written reflection was requested in which students described what they learned, the challenges they faced, and how they solved problems during the activity. The results showed a high level of understanding of the photochemical principles involved and an active participation in all stages of the process.

Integration of Educational Tools

In this study, various educational tools were employed to enrich student learning during the synthesis of paracetamol via free radicals. These tools included a logbook, a pre-exam, a report, and a reflection questionnaire, each contributing uniquely to the pedagogical process.

Logbook. Students completed a logbook before the experiment. This resource allowed them to document their observations and track both individual and group progress. The logbook served as a space for students to express their doubts and consolidate their learning.

Pre-Exam. Before beginning the synthesis, a pre-exam was administered to evaluate the students' initial knowledge of the fundamental chemistry concepts involved. This initial assessment helped identify areas that needed reinforcement and prepared students for the experimental phase.

Report. After the practice, students prepared a structured report in which they detailed their findings and conclusions. This exercise not only reinforced their written communication and critical analysis skills but also allowed for an objective evaluation of their understanding of free-radical synthesis and its applications.

Reflection Questionnaire. Finally, the reflection questionnaire encouraged students to think about their experience and apply what was taught in theory to practice. This tool facilitated self-critique, helping them identify strengths and areas for improvement.

Challenges Faced by Students. Understanding the reaction, being an activity designed for second-semester organic chemistry students, some students had difficulty grasping what free radicals are, which required academic advising to reinforce their knowledge.

Some students struggled with the process of rationalizing and proposing new reaction pathways since this required creative thinking, along with a solid foundation in organic chemistry.

Limitations of the Experimental Activity. Despite the good performance achieved by the students, there was some

variability in the results among teams, which could be attributed to differences in experimental precision, material handling, and interpretation of the procedural steps.

Coping Strategies. Students were encouraged to use literature research to overcome challenges and propose solutions to identified limitations, thus fostering deeper and more collaborative learning.

Practical Tips: Organized Additional Sessions. Academic advising was held for students who needed to reinforce their concepts.

Promoted Teamwork. Encouraged collaboration and idea exchange among students.

Bibliographic Resources. Students were encouraged to use the academic literature to expand their knowledge and skills. Most of these sources are included in the bibliography of this work.

This study was designed to enhance student learning through the development of key competencies, such as chemical reasoning, experimental design, data analysis, and collaborative work. The assessment of learning focused on these competencies, utilizing rubrics to evaluate performance in laboratory work, the quality of experimental design, the analysis of results, and the final report. This methodology ensured an objective and structured evaluation.

This study involved student participation solely for academic purposes as part of their training as future engineers. The activities conducted were regular laboratory practices within an educational context. According to our institutional guidelines, projects of this nature are exempt from ethical review or approval by the Ethics Committee or the Institutional Review Board (IRB).

CONCLUSIONS

The innovative methodology presented in this work, which involves the synthesis of paracetamol through free radicals, represents a pioneering approach that has not been previously reported in the scientific literature. This novel procedure has generated considerable interest among second-semester organic chemistry students, offering them the opportunity to explore a new synthetic route to produce paracetamol, a compound of great importance in the pharmaceutical industry. Throughout the experiment, students integrated spectroscopic techniques such as nuclear magnetic resonance (NMR), mass spectrometry, and melting point determination. The practical application of these analytical methods within the context of an innovative synthesis not only enhanced their technical skills but also deepened their understanding of the fundamental principles of organic chemistry. Additionally, a significant pedagogical aspect that emerged during the experiment was the concept of selectivity in free-radical reactions. Contrary to the common perception that radical reactions are nonselective, students learned that selectivity can, in fact, be achieved through a careful understanding of reaction mechanisms, reagents, and experimental conditions. The results demonstrate that students were highly engaged with the challenge of improving an established synthetic procedure for a compound with health-related applications. This resulted in a solid grasp of theoretical concepts, excellent experimental performance, and well-crafted activity reports. In summary, this synthesis was a great success for the students, providing them with a rich and stimulating educational experience and equipping them with both practical and theoretical tools that reinforce their learning and professional development in organic chemistry.

The objective was achieved: to obtain paracetamol quickly and easily using a methodology that had not previously been reported in the literature. The success of this approach highlights its potential as a valuable addition to the chemistry curriculum, fostering creativity, critical thinking, and technical competence among future chemists.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at <https://pubs.acs.org/doi/10.1021/acs.jchemed.5c00563>.

Supplemental figures, including NMR spectra (Figures S1–S4) and HRMS spectra (Figures S5 and S6) (PDF)
Post-practicum evaluation results, Group 2AM2 (PDF)
Post-practicum evaluation results, Group 2LM2 (PDF)
Logbook, Group 2AM2 (PDF)
Logbook, Group 2LM2 (PDF)
Pre-exam, Group 2AM2 (PDF)
Blank Pre-exam (PDF)
Reflection questionnaire, Group 2AM2 (PDF)
Reflection questionnaire, Group 2LM2 (PDF)
Laboratory report, Group 2AM2, Work team 03 (PDF)
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Notes

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